

mentation, and majority of options are self-explanatory (for example, resolution of pictures/videos).

Open *motion.conf* file in a text editor using the following command (you may copy *motion.conf* file provided with this article):

```
$ sudo nano /home/pi/motion/
motion.conf
```

Make changes to the configuration file such that it can capture videos, save these in media folder and email these to the pre-defined email address. Given below is the list of parameters to be configured in *motion.conf* file.

```
target_dir /home/pi/motion/
media [by default is "target_dir /
tmp/motion" ]
```

```
movie_filename myvideo [by
default is "movie_filename %v-
%Y%m%d%H%M%S"]
```

```
on_movie_end python Send-
MyVideo.py %f [by default is
"on_movie_end value"]
```

In the above configuration, we have used Python script to send emails, which get executed when the captured movie ends.

Script *SendMyVideo.py* is a Python script that is used to send emails with movies as attachments. Copy the script in */home/pi/motion* folder.

Make necessary changes in Python script. Change the text given below with your email credentials, which you will use to send emails:

```
USERNAME = "example@test.
com"
```

```
PASSWORD = "examplepass-
word"
```

Change the email ID in *send-Mail()* ("receiver's email ID here") with the ID of the person who will be receiving the video clips.

Note. In the Python script we have used Gmail as the SMTP server for sending emails. You can make necessary changes to the script in the SMTP section if you want to use any other SMTP server.

Configuration of the system is now complete. Start the world

webcam by simply navigating to the directory and starting Motion using the command given below:

```
$ sudo cd /home/pi/motion
and type
```

```
$ sudo motion
```

Motion takes the local configuration file for execution. Videos will be saved in the sub-folder media. In addition, emails will be sent as configured with attached videos (*myvideo*).

Capturing photos and sending email as attachment

We can use the system to send images in place of videos. The script *SendMyPicture.py* is a Python script that is used to send emails with pictures as attachments. Copy the script in */home/pi/motion* folder.

Make necessary changes in Python script for it to work in your environment. Change the email credentials as was done for *SendMyVideo.py* script for sending videos.

Make necessary changes in *motion.conf* file as given below:

```
Search for #on_picture_save
```

```
python SendMyPicture.py
```

```
Uncomment it by removing #
```

```
Comment on_movie_end python
```

```
SendMyVideo.py %f by adding #
```

```
Change command snapshot_file-
```

```
name %v-%Y%m%d%H%M%S-
```

```
snapshot with lastsnap %v-
```

```
%Y%m%d%H%M%S-snapshot
```

Start Motion using the command

```
$ sudo motion by going to the
/home/pi/motion
```

With the above configurations, Raspi will be configured to send videos and images, which are being captured by the webcam by sensing motion.

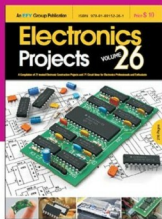
Scripts can be modified to do different tasks—we leave that for you to experiment with! **EFY**



Biswajit Das was manager-
R&D, EFY Labs till recently

Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

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